

Sales Force Automation (SFA)

Master Technical & Operational Documentation | Complete System Architecture & Reference Guide

Sales Force Automation (SFA) — Project Master Documentation

NOTE: This master document provides a technical and operational overview of the **Sales Force Automation (SFA)** platform. It covers system architecture, role-based access control (RBAC), end-to-end business workflows, database models, target/performance frameworks, and analytics modules.

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1. Executive Summary & System Purpose

The **Sales Force Automation (SFA)** platform is an operational-sales management system designed for multi-branch distribution, field force tracking, inventory logistics, and order fulfillment.

IMPORTANT: Business Focus Directive: SFA is an **operational-sales system**, not a financial accounting software. Primary performance indicators focus on operational throughput (Tasks Completed, Orders Fulfilled, Workforce Efficiency) alongside real-time sales order revenue tracking.

Key Capabilities

- **Multi-Tenant & Multi-Branch Architecture:** Isolated data scopes per Organization, Branch, Warehouse, and Team.
- **Field Force Automation:** Real-time task tracking, customer visit logging, digital signature capture, and geofence-assisted handovers.
- **Warehouse & Inventory Integration:** Multi-warehouse stock tracking, stock movements, reserved vs. available quantities, and automated handover approvals.
- **Operational Target & Performance System:** Dynamic operational performance tracking for Sales Executives (Task Execution %) and Sales Managers (Order Fulfillment %).
- **Visualization-First Reports:** 0-export interactive Recharts analytics describing Branch Performance, Field Force Workforce, Warehouse Stock, and Sales Revenue Trajectory.

2. Technology Stack & System Architecture

```
graph TD
  Client[React Frontend - Vite + TailwindCSS + Recharts] -->|REST API + JWT| Gateway[Express.js Backend API]
  Gateway --> Middleware[Auth & RBAC Middleware]
  Middleware --> Services[Module Business Logic Services]
  Services --> Repos[Repository Data Access Layer]
  Repos --> Prisma[Prisma ORM]
  Prisma --> DB[(PostgreSQL Database - Neon/Managed PG)]
  Services --> Storage[Local File Storage - Avatars & Signatures]
```

Stack Breakdown

- **Frontend:** React (Vite), TailwindCSS, Recharts for dynamic visual graphs, Lucide React icons, Framer Motion for micro-animations.
- **Backend:** Node.js, Express.js (Modular Monolith Architecture), Prisma ORM (Client v7+).
- **Database:** PostgreSQL (Hosted on Neon / Managed Postgres) with SSL pooling.
- **Authentication:** JWT-based Bearer Authentication with password hashing via bcryptjs.
- **File Storage:** Local uploads directory with static express routing for user avatars and digital signatures.

3. Role-Based Access Control (RBAC) Matrix

The system enforces strict role-based scoping across 7 distinct roles:

Role Name	Access Level	Primary System Focus
Organization Super Admin	Platform-Wide	Global organization oversight, Company Admin management, multi-branch analytics.
Company Admin	Organization Level	Branch, Department, Team, and User account administration.
Head of Sales	Strategic Sales Level	Revenue trajectory, Target & Performance dashboard, company-wide operational reports.
Sales Manager	Branch Level	Branch operational management, team task assignment, branch sales order completion.
Sales Executive	Field / Individual Level	Task execution, customer visits, digital signature capture, sales order placement.
Inventory Manager	Global Logistics	Product catalog, multi-warehouse stock management, global stock transfers.
Warehouse Manager	Warehouse Console	Warehouse stock check-in/check-out, pickup handover approvals, stock movements.

4. Core Business Workflows & Lifecycles

A. Sales Order Lifecycle

```
stateDiagram-v2
    [*] --> DRAFT: Order Created
    DRAFT --> SUBMITTED: Submitted by Executive/Manager
    SUBMITTED --> APPROVED: Manager Approval
    APPROVED --> IN_FULFILLMENT: Assigned for Pickup/Delivery
    IN_FULFILLMENT --> DELIVERED: Customer Handover Complete
    DELIVERED --> COMPLETED: Handover Approved & Invoiced
    DELIVERED --> CANCELLED: Order Cancelled
```

- **Order Creation (^DRAFT^):** Created by Sales Executive or Sales Manager with selected customer, line items, quantities, and prices.
- **Order Submission (^SUBMITTED^):** Order lock for approval review.

- **Approval & Allocation (`APPROVED`)**: Verified against stock availability in designated warehouses.
- **Fulfillment & Delivery (`IN\_FULFILLMENT` \$\rightarrow\$ `DELIVERED`)**: Field executive picks up items and delivers to customer.
- **Invoice & Completion (`COMPLETED`)**: Customer signs digital delivery proof; Warehouse Manager approves handover, automatically triggering stock deduction and invoice generation.

B. Task Execution & Warehouse Handover Flow

```
sequenceDiagram
    autonumber
    actor Manager as Sales Manager
    actor Worker as Field Sales Executive
    actor WM as Warehouse Manager

    Manager->>Worker: Assign Delivery Task with Sales Order
    Worker->>WM: Arrive at Warehouse for Stock Pickup
    WM-->>Worker: Release Products for Delivery
    Worker->>Customer: Complete Delivery & Capture Digital Signature
    Worker->>WM: Submit Handover Proof (Signature + Photos)
    WM->>WM: Approve Handover Verification
    Note over WM: System deducts stock & generates completed invoice
```

C. Inventory & Stock Movement Flow

- **Net Available Quantity Calculation:**

$$\text{Net Available Qty} = \text{Total Physical Quantity} - \text{Reserved Quantity}$$

- **Stock Movement Types:**

- `CHECK\_IN`: Initial stock load or purchase receipt into warehouse. - `CHECK\_OUT`: Manual stock adjustment or write-off. - `TRANSFER\_IN` / `TRANSFER\_OUT`: Inter-warehouse or branch stock transfers. - `FULFILLMENT\_DEDUCTION`: Automated stock deduction upon completed sales order handover. ---

5. Target & Operational Performance Module

The **Target & Performance** module evaluates users strictly by **Operational Metrics** appropriate to their role:

```
graph LR
    subgraph Sales_Executive_Target [Sales Executive Target]
        SE_A[Tasks Assigned] --> SE_B[Tasks Completed]
        SE_B --> SE_C[Task Execution Rate %]
    end
```

```

end

subgraph Sales Manager Target
  SM_A[Sales Orders Received] --> SM_B[Sales Orders Completed]
  SM_B --> SM_C[Order Fulfillment Rate %]
end

subgraph Overall Organization Graph
  ORG_A[Sales Orders Revenue] --> ORG_B[Daily / Weekly / Monthly Revenue Trajectory]
end

```

Role-Based Performance Metrics

- **Sales Executives:**
 - **Target Indicator: Tasks Assigned vs Tasks Completed.** - **Formula:** $\frac{\text{Task Execution \%}}{\frac{\text{Completed Tasks}}{\text{Total Assigned Tasks}}} \times 100\%$
- **Sales Managers:**
 - **Target Indicator: Sales Orders Placed/Received vs Sales Orders Completed.** - **Formula:** $\frac{\text{Order Completion \%}}{\frac{\text{Completed Orders}}{\text{Total Orders Placed}}} \times 100\%$
- **Organization Revenue Trajectory:**
 - Dynamic Area & Line charts plotting real Sales Order Revenue (₹) over **Daily, Weekly, and Monthly** periods. --

6. Reports & Business Intelligence Module

The **Reports Page** is designed with **0 export/download buttons** to provide an immediate, visual-first executive dashboard across 5 core tabs:

- **Branch Operational Performance:**
 - Real DB counts for `Total Orders`, `Completed Tasks`, `Requested Quantity`, `Staff Count` per branch.
- **Warehouses Report:**
 - Total warehouses, capacity utilization %, total stock value, and warehouse manager assignments.
- **Product Report:**
 - Product catalog metrics, price list, category breakdown, and total quantities sold.
- **Stock per Branches Report:**
 - Table detailing Product, SKU, Warehouse, Branch Assignment, and **Net Available Quantity**.
- **Field Force Workforce & Task Execution Report:**
 - Field workforce count, overall task completion rate %, and per-worker task breakdown table (Worker Name, Email, Branch, Assigned Tasks, Completed Tasks, Pending Tasks). ---

7. Database Schema & Data Models

Core Entities & Relationships

```
erDiagram
    ORGANIZATION ||--|{ BRANCH : owns
    ORGANIZATION ||--|{ WAREHOUSE : owns
    ORGANIZATION ||--|{ USER : employs
    BRANCH ||--|{ USER : contains
    WAREHOUSE ||--|{ STOCK : stores
    PRODUCT ||--|{ STOCK : tracked_in
    USER ||--|{ ORDER : places
    USER ||--|{ TASK : assigned_to
    CUSTOMER ||--|{ ORDER : receives
    ORDER ||--|{ ORDER_ITEM : contains
    PRODUCT ||--|{ ORDER_ITEM : ordered_in
```

Primary Models (`prisma/schema.prisma`)

- **`Organization`**: Multi-tenant root entity.
- **`Branch`**: Physical business locations linked to users, orders, and warehouses.
- **`User`**: Account entity linked to `UserRole`, `Branch`, `Department`, `Team`. Includes `avatarUrl` for profile storage.
- **`Warehouse` & `Stock`**: Logistics inventory entities tracking physical quantity, reserved quantity, and location coordinates.
- **`Order` & `OrderItem`**: Sales order entities tracking customer, owner, status, line items, and total amount.
- **`Task`**: Field delivery task linked to assigned worker, pickup location, delivery location, customer signature, and status (`PENDING`, `IN\_PROGRESS`, `COMPLETED`).
- **`Target`**: Operational performance target records for users and teams.

8. API Architecture & Route Sitemap

API Endpoints Overview

Base Endpoint	Method	Description
`/api/v1/auth/login`	`POST`	User login & JWT issuance
`/api/v1/auth/me`	`GET`	Current authenticated user profile
`/api/v1/users`	`GET`, `POST`	User management (Supports `limit=1000` for full org list)
`/api/v1/users/profile`	`PUT`	Update user profile & avatar upload
`/api/v1/sales-orders`	`GET`, `POST`	Sales order CRUD & lifecycle updates
`/api/v1/target-performance/company-overview`	`GET`	Target & performance operational analytics & trend graphs
`/api/v1/reports/organization-analytics`	`GET`	5-tab operational reports payload
`/api/v1/inventory/warehouses`	`GET`, `POST`	Warehouse management
`/api/v1/inventory/stock`	`GET`, `POST`	Stock level tracking & movements
`/api/v1/team/tasks`	`GET`, `POST`	Task creation, status updates & signature handover

9. User Settings & Profile Persistence

Avatar / Profile Picture Flow

- **Upload:** User selects image in Settings Page (`SettingsPage.jsx`).
- **Storage:** Backend saves file to `/uploads/avatars/avatar-[userId]-[timestamp].[ext]`.
- **Database:** Updates `User.avatarUrl` column in PostgreSQL database.
- **Auth Sync:** `AuthContext.jsx` updates local storage state so the avatar persists seamlessly across page refreshes and sessions.

TIP: Verification Status: All modules described in this documentation have been verified against the live PostgreSQL database and production Vite build (`npm run build` completed in **1.94s** with 0 errors).